

FIXORA

Next-Gen Doorstep Home Service Engine

An Enterprise MERN Stack Web Platform with Auto-Priority Ranking, Satellite GPS Pin-Dropping, and Automated Cashflow Billing

React.js (Vite)

Frontend UI Engine

Node.js + Express

REST API Backend

MongoDB

Database Layer

Google Maps + UPI QR

External Integrations

Traditional Challenges »→ Fixora Solutions

Traditional Challenges

Address Navigation Friction

Technicians face delays due to inaccurate address inputs and manual navigation, causing significant service time loss.

Manual Job Allocation

Unorganized, human-dependent job dispatching leads to mismatches, inefficiencies, and delayed expert assignment.

Opaque Spare Parts Pricing

Customers receive unclear billing with no itemized breakdown, eroding trust and creating billing disputes.

Fixora Solutions

Satellite Live GPS Pin-Drop

Extracts exact satellite GPS coordinates (lat, lng) for precise, real-time location sharing with technicians and admins.

Smart Auto-Select Allocation

System intelligently auto-routes job requests to the correct expert, enabling zero-friction admin dispatching.

Automated GST & UPI Invoicing

Generates itemized invoices with instant dynamic UPI QR code for scan-to-pay cashless billing on every job.

Full-Stack MERN Technical Architecture

MERN STACK ARCHITECTURE

Vertical Data Flow & Architecture

1. FRONTEND CLIENT LAYER

React.js (Vite) + Bootstrap 5 UI + Axios Interceptors

React.js (Vite Framework)

Bootstrap 5 UI Components

Axios Interceptors

HTML5 Geolocation API

REST API (JSON Data)

2. BACKEND API LAYER

Node.js Runtime + Express.js REST APIs

Express REST API Endpoints

JWT Auth & Middleware Guards

Role-Based Access Control (RBAC)

Controller Business Logic

Mongoose Queries / ODM

3. DATABASE & EXTERNAL SERVICES

MongoDB Atlas + Cloudinary + Google Maps

MongoDB (Mongoose ODM)

Cloudinary (Repair Photos)

Google Maps Embed API

Dynamic UPI QR Engine

* Full-Stack MERN Architecture with Encrypted JWT Security & REST Endpoints

Frontend Layer

- React.js (Vite) — lightning-fast HMR dev builds
- Modular State Management & Axios Interceptors
- Bootstrap 5 responsive UI components

Backend API Layer

- Node.js & Express RESTful API endpoints
- protect & authorize JWT Middleware guards
- Role-based access: Admin, Technician, Customer

Database Layer

- MongoDB with Mongoose ODM schemas
- Relational .populate() references across collections
- Cascading delete handling for data integrity

External Integrations

- Cloudinary API — repair photo upload & storage
- Google Maps Embed API — live map previews
- Dynamic UPI QR — scan-to-pay invoice payloads

Priority Ranking Engine

Subscription Tier Logic — **Higher Plan = Top Algorithmic Placement & Auto-Assigned Initial Rating.** MongoDB sorts on `planPrice: -1`, `rating: -1`.

- **Platinum Tier ($\geq ₹5,000$)**
 - ★ 4.9 Stars — Rank #1 Top Priority Placement. Highest algorithmic weight in customer-facing service catalog.
- **Gold Tier (₹3,000 – ₹4,999)**
 - ★ 4.7 Stars — Rank #2 Priority Placement. Strong visibility with premium listing position.
- **Silver Tier (₹1,500 – ₹2,999)**
 - ★ 4.5 Stars — Rank #3 Priority Placement. Mid-tier visibility with competitive rating baseline.
- **Basic Tier ($< ₹1,500$)**
 - ★ 4.3 Stars — Standard Placement. Entry-level listing with organic search ranking only.

PRIORITY RANKING ENGINE

Algorithmic Subscription Tier Hierarchy

RANK #1

PLATINUM TIER

Subscription Investment: $\geq ₹5,000$ / mo

4.9 ★★★★★

Top Priority Visibility | Auto Rating 4.9

RANK #2

GOLD TIER

Subscription Investment: ₹3,000 – ₹4,999

4.7 ★★★★★

High Priority Placement | Auto Rating 4.7

RANK #3

SILVER TIER

Subscription Investment: ₹1,500 – ₹2,999

4.5 ★★★★★

Competitive Placement | Auto Rating 4.5

RANK #4

BASIC TIER

Subscription Investment: $< ₹1,500$

4.3 ★★★★★

Standard Organic Listing | Auto Rating 4.3

● MongoDB Sorting Engine: { planPrice: -1, rating: -1 }

Satellite Live GPS Geolocation & Map Gateway

Precision location technology embedded directly into the booking workflow — no manual address entry required.

Native HTML5 GPS Pin-Drop

Customer taps once to extract exact browser satellite GPS coordinates (lat, lng) — eliminating address input errors entirely.

Interactive Map Preview

Embedded real-time Google Maps frame renders inside the booking modal, giving customers instant visual confirmation of their pin location.

Admin Dashboard Link

Clickable "Open GPS Map Location" hyperlink auto-generated on Admin dashboard for every booking — one-click navigation dispatch.

Technician Dashboard Link

Same direct GPS navigation hyperlink pushed to the Technician portal — technician opens map and drives straight to customer site.

Customer Portal & Real-Time Status Tracker

Smart Service Catalog

AC

Refrigerator

Washing Machine

RO Purifier

Electrical

Plumbing

Dedicated appliance categories with expert technician routing per service type.

Real-Time 4-Step Tracker Bar

01

Request Received

Booking confirmed, GPS coordinates captured, service ticket created in MongoDB.

02

Expert Assigned

Admin reviews auto-selected technician and dispatches with one click.

03

Tech Accepted / On Way

Technician accepts job via portal; customer status updates in real time.

04

Resolved & Billed

Job closed, invoice auto-generated with UPI QR, customer prompted for 5-star feedback.

Smart Auto-Select Allocation Engine

Intelligent request routing that eliminates manual admin lookup and enables zero-click technician dispatching.

1

Requested Tech Auto-Routing

When a customer requests a specific expert (e.g., Zayan or Aarav), the system extracts `requestedTechId` and pre-selects them in the Admin Allocation dropdown — no manual search needed.

2

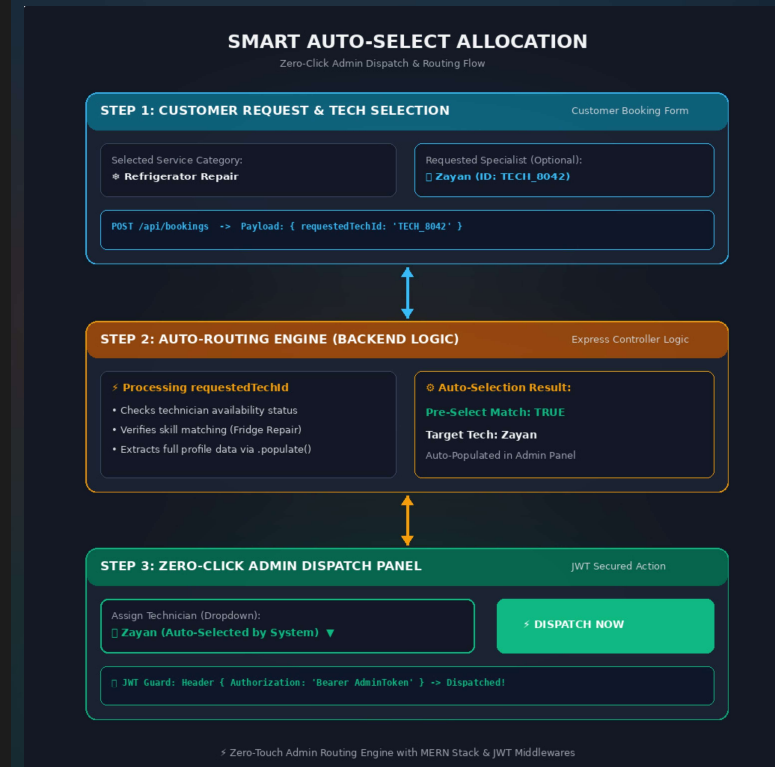
Zero-Click Admin Efficiency

Admin simply reviews the auto-selected technician suggestion and clicks "Allocate" once to dispatch instantly — full job assignment in a single action.

3

JWT-Secured Dispatch

All allocation API calls pass through `protect` & `authorize` JWT middleware, ensuring only authenticated admins can dispatch technicians.





TECHNICIAN PORTAL

Technician Portal & Work Verification Log



Duty Management Queue

Real-time Job Accept and Reject controls with JWT authorization headers on every API call — technicians only see their own assigned jobs.



3-Stage Work Inspection Proofs

1. Customer Appliance Issue Photo → 2. Before-Repair Proof Photo → 3. After-Repair Fixed Photo. All uploads routed through Cloudinary API.



Dynamic Spare Parts Input

Add or remove spare part rows dynamically with clean price-erase validation — each row feeds directly into the automated billing formula.



JWT Role Authorization

Technician-scoped JWT tokens restrict portal access — technicians cannot view admin panels, customer PII, or other technicians' job queues.

Automated Billing & Dynamic UPI QR System

Automated Cost Formula

**Total Bill = ₹350 Visiting Charge
+ $\Sigma(\text{Spare Part Qty} \times \text{Item Price})$**
— computed server-side with zero
manual input required.

Dynamic UPI QR Code

Generates scan-to-pay QR
dynamically on the PDF invoice
using payload:

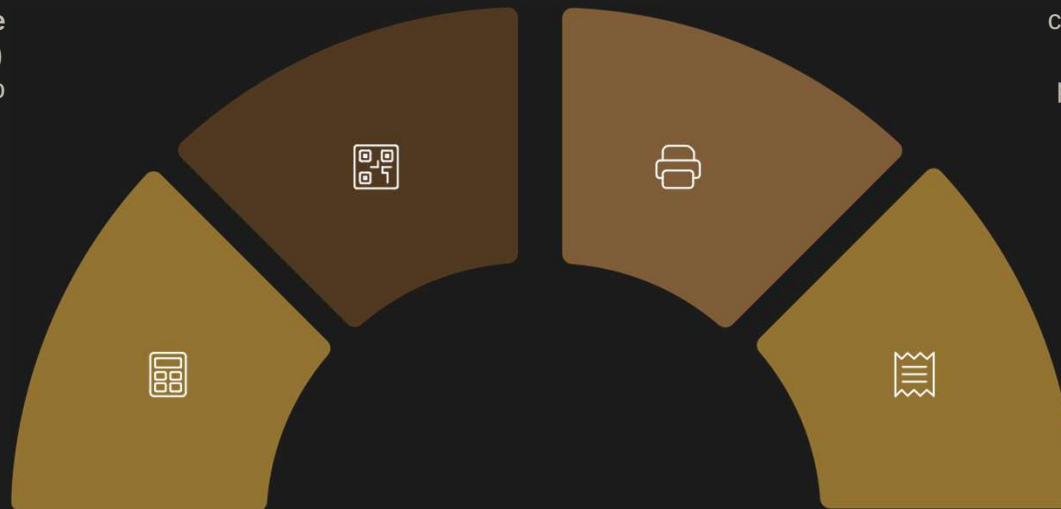
`upi://pay?pa=fixora@upi&am=TO
TAL_AMOUNT` — auto-populated per
job total.

Print-Ready Invoicing

Printable A4 layout with GST-
compliant itemized breakdown and
payment mode toggle — Cash vs.
UPI — selectable before printing.

GST & Itemized Breakdown

Every invoice auto-populates
customer details, technician info,
spare parts list, quantities, unit
prices, and applicable GST line
items.



Security, Future Scope & Wrap-Up

FIXORA delivers a production-ready, highly secure, and fully functional MERN stack solution.



Enterprise Security

JWT Token headers on all protected endpoints, Mongoose `returnDocument: 'after'` compliance, and atomic database rollbacks for data integrity.



Future Scaling Roadmap

WhatsApp Webhook SMS alerts, React Native Mobile App for Field Technicians, and Razorpay Gateway Checkout integration planned for v2.



Thank You — Q&A

FIXORA is a complete, enterprise-grade home service engine. Open for questions on architecture, security design, or any implementation detail.